Chapter 9

Introduction to Flight Dynamics

This chapter will treat the key concepts and formulations used in the discipline of flight dynamics and control. The primary focus here will be on the aerodynamic characterization of the aircraft, as required for flight dynamics applications. For a complete treatment of the subject, see Etkin [65] and Nelson [66].

9.1 Frames of Reference

Description of body or fluid positions, velocities, and rotation rates requires a frame of reference for these quantities. For unsteady aerodynamics and flight dynamics, two distinct frames of reference are useful:

- An inertial "Earth" frame, either fixed to the Earth or translating uniformly relative to the Earth.
- The non-inertial "body" frame, typically fixed to the aerodynamic body of interest.

Figure 9.1 illustrates the distinction between the two frames. For steady aerodynamics the body frame is most natural, although the Earth (or airmass) frame is also useful as in the case of Trefftz-plane theory treated in Chapter 5. For unsteady aerodynamics the Earth frame is more natural because it is inertial.

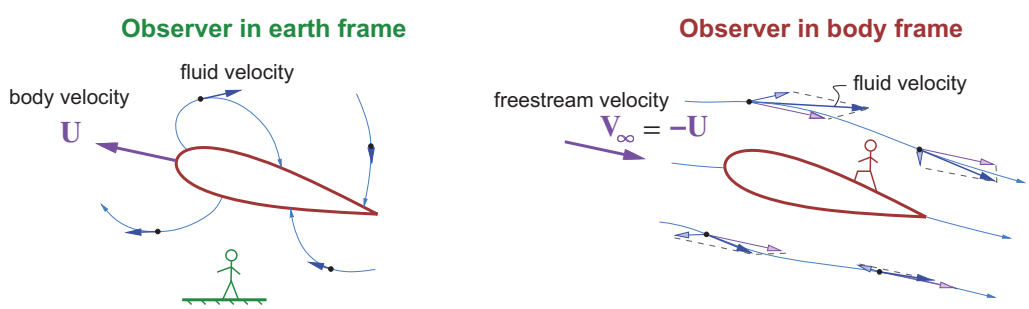

Figure 9.1: Body moving at velocity $\mathbf{U}$ through fluid, as seen in the Earth and body frames.

9.2 Axis Systems

The orientation of the axis system used to quantify any vector or tensor is arbitrary. Common choices are to align one axis (e.g. the x axis) with either a body feature such as an airfoil chordline, or with the freestream direction. If the body is only translating, a common axis system can be used for all quantities. However, if the body is rotating, then two axis systems naturally arise:

- "Earth" axes x^e, y^e, z^e fixed to the ground. These are non-rotating (the Earth's rotation is neglected).
- "Body" axes x^b, y^b, z^b fixed to the body. These rotate along with the body relative to inertial space.

Note that "frame" and "axes" are distinct concepts. For example, the aircraft position, velocity, and rotation rate $\mathbf{R}_o, \mathbf{U}, \mathbf{\Omega}$ treated in Chapter 7 were all defined in the Earth frame. But $\mathbf{R}_o$ is usually expressed via its Earth-axes x^e, y^e, z^e components, while $\mathbf{U}, \mathbf{\Omega}$ are usually expressed via the aircraft's body-axes x^b, y^b, z^b components. Relating them requires the axis transformation relations developed in Appendix F.

9.3 Body Position and Rate Parameters

Application of the equations of fluid motion in unsteady or quasi-steady flow situations requires specification of the velocity of every point on the body surface. This is parameterized by the inertial-frame quantities,

$$\begin{array}{ll} \mathbf{U}(t) & \text{velocity vector of body reference point O} \\ \mathbf{\Omega}(t) & \text{rotation-rate vector of body} \end{array}$$

so that a point P at some location $\mathbf{r}_\mathrm{p}$ on the body relative to point O has the following velocity.

$$\mathbf{U}_\mathrm{p} \;=\; \mathbf{U} + \mathbf{\Omega}\times\mathbf{r}_\mathrm{p}$$

In aircraft flight dynamics applications the atmosphere is occasionally specified to have some nonuniform "gust" velocity field $\mathbf{V}_\mathrm{gust}(\mathbf{R},t)$, whose effect on the forces on the aircraft is to be determined. To evaluate this gust velocity at each point $\mathbf{r}_\mathrm{p}$ on the aircraft requires knowing that point's Earth position within the gust field. This is parameterized by

$$\begin{array}{ll} \mathbf{R}_\mathrm{o}(t) & \text{position of body reference point O} \\ \phi,\theta,\psi(t) & \text{body roll, elevation, and heading Euler angles} \end{array}$$

which are sketched in Figure 9.2. The Earth position of point P and the gust velocity at that point can then be obtained from the $\mathbf{V}_\mathrm{gust}(\mathbf{R},t)$ field.

$$\mathbf{R}_\mathrm{p}(\mathbf{r},t) \;=\; \mathbf{R}_\mathrm{o}(t) + \mathbf{r}_\mathrm{p} \tag{9.1}$$

$$(\mathbf{V}_\mathrm{gust})_\mathrm{p} \;=\; \mathbf{V}_\mathrm{gust}(\mathbf{R}_\mathrm{p},t) \tag{9.2}$$

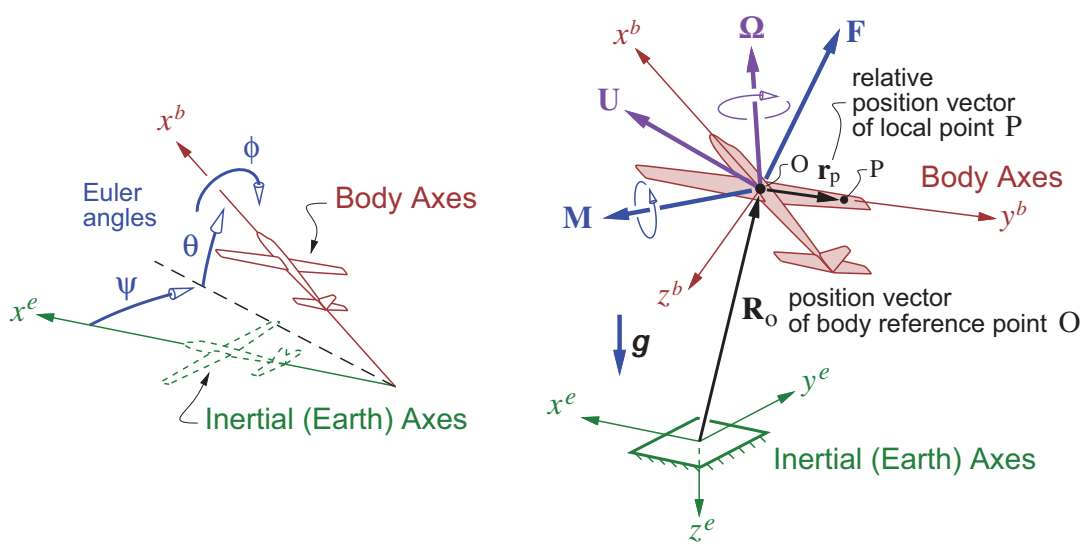

Figure 9.2: Aircraft motion parameters. Total aerodynamic force $\mathbf{F}$ and moment $\mathbf{M}$ are also shown. Euler angles define transformation matrices $\bar{\bar{\mathbf{T}}}_b^e$ or $\bar{\bar{\mathbf{T}}}_e^b$ between body and Earth axes, developed in Appendix F.

9.4 Axis Parameterization and Conventions

The body reference-point position vector $\mathbf{R}_\mathrm{o}$ is best given by its components along the inertial Earth axes, shown as x^e, y^e, z^e in Figure 9.2. In contrast, the local position vector $\mathbf{r}$, and the body velocity and rotation

rate vectors $\mathbf{U}, \mathbf{\Omega}$ are best given in the body axes x^b, y^b, z^b, since these are also used to describe the flow about the body, e.g. via the velocity field $\mathbf{V}(\mathbf{r},t)$. To translate the individual components of any vector, such as $\mathbf{U}$, from body to Earth axes, we apply the general vector transformation (F.1) derived in Appendix F.

$$\mathbf{U}^e \equiv \begin{Bmatrix} U_x^e \\ U_y^e \\ U_z^e \end{Bmatrix} = \begin{bmatrix} & & \\ & \bar{\bar{\mathbf{T}}}_b^e & \\ & & \end{bmatrix} \begin{Bmatrix} U_x^b \\ U_y^b \\ U_z^b \end{Bmatrix} \equiv \bar{\bar{\mathbf{T}}}_b^e \mathbf{U}^b \tag{9.3}$$

The direction-cosine transformation matrix $\bar{\bar{\mathbf{T}}}_b^e$ is now formed as the product of three simple rotation matrices for the individual Euler angles in the standard sequence $-\phi, -\theta, -\psi$:

$$\begin{aligned} \bar{\bar{\mathbf{T}}}_b^e &= \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi & \cos\phi \end{bmatrix} \\ &= \begin{bmatrix} \cos\theta\,\cos\psi & \sin\phi\,\sin\theta\,\cos\psi - \cos\phi\,\sin\psi & \cos\phi\,\sin\theta\,\cos\psi + \sin\phi\,\sin\psi \\ \cos\theta\,\sin\psi & \sin\phi\,\sin\theta\,\sin\psi + \cos\phi\,\cos\psi & \cos\phi\,\sin\theta\,\sin\psi - \sin\phi\,\cos\psi \\ -\sin\theta & \sin\phi\,\cos\theta & \cos\phi\,\cos\theta \end{bmatrix} \end{aligned} \tag{9.4}$$

The reciprocal conversion matrix $\bar{\bar{\mathbf{T}}}_e^b$ is composed of the reverse rotation sequence ψ, θ, ϕ. But this is also the inverse of $\bar{\bar{\mathbf{T}}}_b^e$, which is simply its transpose as derived in Appendix F for the general case.

$$\bar{\bar{\mathbf{T}}}_e^b = \bar{\bar{\mathbf{T}}}_b^{e^{-1}} = \bar{\bar{\mathbf{T}}}_b^{e^T} \tag{9.5}$$

As an application example, consider equations (9.1) and (9.2) used to obtain the gust velocity at an aircraft point P. The aircraft position $\mathbf{R}_\mathrm{o}$ and the $\mathbf{V}_\mathrm{gust}(\mathbf{R},t)$ function are typically provided in Earth axes, i.e. as $\mathbf{R}_\mathrm{o}^e$ and $\mathbf{V}_\mathrm{gust}^e(\mathbf{R}^e,t)$, while the point P vector $\mathbf{r}_\mathrm{p}$ is known in body axes, as $\mathbf{r}_\mathrm{p}^b$. Expressions (9.1) and (9.2) would therefore need to be evaluated as

$$\mathbf{R}_\mathrm{p}^e = \mathbf{R}_\mathrm{o}^e + \bar{\bar{\mathbf{T}}}_b^e \mathbf{r}_\mathrm{p}^b \tag{9.6}$$

$$\mathbf{V}_{\mathrm{gust}_\mathrm{p}}^b = \bar{\bar{\mathbf{T}}}_e^b \mathbf{V}_\mathrm{gust}^e(\mathbf{R}_\mathrm{p}^e,t) \tag{9.7}$$

with the final result $\mathbf{V}_{\mathrm{gust}_\mathrm{p}}^b$ being in the body axes. This would then be usable for calculation of body forces and moments which are typically performed in the body axes.

9.5 Flow Angles

The aircraft's velocity $\mathbf{U}$ components in the body axes can be defined in terms of its magnitude V_∞ and the angles of attack α and sideslip angle β, or vice versa. In the standard convention these relations are

$$\mathbf{U}^b = \begin{Bmatrix} U_x^b \\ U_y^b \\ U_z^b \end{Bmatrix} = -\mathbf{V}_\infty^b = V_\infty \begin{Bmatrix} \cos\alpha\,\cos\beta \\ \sin\beta \\ \sin\alpha\,\cos\beta \end{Bmatrix} \tag{9.8}$$

$$\begin{aligned} V_\infty &= \sqrt{(U_x^b)^2 + (U_y^b)^2 + (U_z^b)^2} \\ \alpha &= \arctan\left(U_z^b / U_x^b\right) \\ \beta &= \arctan\left(U_y^b / \sqrt{(U_x^b)^2 + (U_z^b)^2}\right) \end{aligned} \tag{9.9}$$

so that $\{U_x^b, U_y^b, U_z^b\}$ and $\{V_\infty, \alpha, \beta\}$ are equivalent alternative parameter sets, related by the reciprocal relations (9.8) and (9.9).

9.6 Aircraft Kinematic Relations

This section derives the kinematic equations of aircraft motion used in flight dynamics and control. See Etkin [65] and Nelson [66] for further details.

9.6.1 Aircraft position rate

The rate of the position $\mathbf{R}_{\mathrm{o}}^{e}$ in Earth axes equals its velocity $\mathbf{U}^{e}$ in Earth axes.

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{R}_{\mathrm{o}}^{e} \;=\; \mathbf{U}^{e} \tag{9.10}$$

In practice this is expressed as

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{R}_{\mathrm{o}}^{e} \;=\; \bar{\bar{\mathbf{T}}}_{b}^{e}\mathbf{U}^{b}} \tag{9.11}$$

since the velocity is defined as $\mathbf{U}^{b}$ in the body axes. Integration of (9.11) to obtain the aircraft trajectory $\mathbf{R}_{\mathrm{o}}(t)$ therefore requires not only the aircraft velocity $\mathbf{U}^{b}(t)$, but also the concurrent Euler angles $\phi, \theta, \psi(t)$ which are needed to compute the transformation matrix $\bar{\bar{\mathbf{T}}}_{b}^{e}(t)$ at each step in time.

9.6.2 Aircraft orientation rate

Appendix F derives the expression (F.7) for the rotation-rate matrix $\bar{\bar{\mathbf{\Omega}}}^{e}$. Choosing the (3,2), (1,3), (2,1) elements of this expression and stacking them in that order as a vector gives a relation between the rotation rate's Earth-axis components and the Euler angle rates.

$$\begin{Bmatrix} \Omega_x^e \\ \Omega_y^e \\ \Omega_z^e \end{Bmatrix} \;=\; \begin{bmatrix} \\ \bar{\bar{\mathbf{K}}}_b^e \\ \\ \end{bmatrix} \begin{Bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{Bmatrix} \tag{9.12}$$

$$\text{where} \qquad \bar{\bar{\mathbf{K}}}_b^e \;=\; \begin{bmatrix} \cos\psi\,\cos\theta & -\sin\psi & 0 \\ \sin\psi\,\cos\theta & \cos\psi & 0 \\ -\sin\theta & 0 & 1 \end{bmatrix} \tag{9.13}$$

Pre-multiplying (9.12) by $\bar{\bar{\mathbf{T}}}_{e}^{b}$ puts this result in body axes.

$$\begin{Bmatrix} \Omega_x^b \\ \Omega_y^b \\ \Omega_z^b \end{Bmatrix} \;=\; \begin{bmatrix} \\ \bar{\bar{\mathbf{T}}}_e^b \\ \\ \end{bmatrix} \begin{Bmatrix} \Omega_x^e \\ \Omega_y^e \\ \Omega_z^e \end{Bmatrix} \;=\; \begin{bmatrix} \\ \bar{\bar{\mathbf{T}}}_e^b \\ \\ \end{bmatrix} \begin{bmatrix} \\ \bar{\bar{\mathbf{K}}}_b^e \\ \\ \end{bmatrix} \begin{Bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{Bmatrix} \tag{9.14}$$

Evaluating the matrix-matrix product above gives

$$\mathbf{\Omega}^{b} \;=\; \begin{Bmatrix} \Omega_x^b \\ \Omega_y^b \\ \Omega_z^b \end{Bmatrix} \;=\; \begin{bmatrix} 1 & 0 & -\sin\theta \\ 0 & \cos\phi & \sin\phi\,\cos\theta \\ 0 & -\sin\phi & \cos\phi\,\cos\theta \end{bmatrix} \begin{Bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{Bmatrix} \tag{9.15}$$

which can then be inverted to give the rate equation for the Euler angles.

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\begin{Bmatrix} \phi \\ \theta \\ \psi \end{Bmatrix} \;=\; \begin{bmatrix} 1 & \sin\phi\,\tan\theta & \cos\phi\,\tan\theta \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi/\cos\theta & \cos\phi/\cos\theta \end{bmatrix} \begin{Bmatrix} \Omega_x^b \\ \Omega_y^b \\ \Omega_z^b \end{Bmatrix}} \tag{9.16}$$

9.7 Dynamics Relations

9.7.1 Linear momentum

The linear momentum equation (Newton's Second Law) for an aircraft, properly expressed in the inertial Earth frame and Earth axes is

$$\mathbf{F}^e + m\,\boldsymbol{g}^e = m\,\frac{\mathrm{d}\mathbf{U}^e}{\mathrm{d}t} \tag{9.17}$$

where the total force acting on the aircraft is the aerodynamic force $\mathbf{F}$ plus the gravity force $m\boldsymbol{g}$, shown in Figure 9.2. The equivalent form in the more convenient body axes is obtained using the transformation matrix, by substituting $\mathbf{F}^e = \bar{\bar{\mathbf{T}}}_b^e\mathbf{F}^b$, etc., and performing a few manipulations as follows.

$$\begin{aligned}
\bar{\bar{\mathbf{T}}}_b^e\,\mathbf{F}^b + m\,\bar{\bar{\mathbf{T}}}_b^e\,\boldsymbol{g}^b &= m\,\frac{\mathrm{d}}{\mathrm{d}t}(\bar{\bar{\mathbf{T}}}_b^e\mathbf{U}^b) \\
&= m\left(\bar{\bar{\mathbf{T}}}_b^e\dot{\mathbf{U}}^b + \dot{\bar{\bar{\mathbf{T}}}}_b^e\mathbf{U}^b\right) \\
&= m\left(\bar{\bar{\mathbf{T}}}_b^e\dot{\mathbf{U}}^b + \vec{\bar{\boldsymbol{\Omega}}}^e\bar{\bar{\mathbf{T}}}_b^e\mathbf{U}^b\right) \\
&= m\left(\bar{\bar{\mathbf{T}}}_b^e\dot{\mathbf{U}}^b + \vec{\bar{\boldsymbol{\Omega}}}^e\mathbf{U}^e\right) \\
&= m\left(\bar{\bar{\mathbf{T}}}_b^e\dot{\mathbf{U}}^b + (\boldsymbol{\Omega}\times\mathbf{U})^e\right)
\end{aligned} \tag{9.18}$$

Relation (F.9) has been used to replace $\dot{\bar{\bar{\mathbf{T}}}}_b^e$ in the third step above. Premultiplying the final result by $\bar{\bar{\mathbf{T}}}_e^b$ gives the linear momentum equation for the aircraft with all quantities specified in the body axes.

$$\boxed{\mathbf{F}^b + m\,\boldsymbol{g}^b = m\left(\dot{\mathbf{U}}^b + \boldsymbol{\Omega}^b\times\mathbf{U}^b\right)} \tag{9.19}$$

9.7.2 Angular momentum

The angular momentum equation for the aircraft is

$$\mathbf{M}^e = \frac{\mathrm{d}\mathbf{H}^e}{\mathrm{d}t} \tag{9.20}$$

$$\mathbf{H}^e = \bar{\bar{\mathbf{I}}}^e\,\boldsymbol{\Omega}^e + \mathbf{h}^e \tag{9.21}$$

where the total angular momentum vector $\mathbf{H}$ has been introduced, with $\mathbf{h}$ being any onboard angular momentum due to propellers, turbines, etc. The total aerodynamic moment $\mathbf{M}$, shown in Figure 9.2, is assumed to be taken about the center of mass, and $\bar{\bar{\mathbf{I}}}$ is the mass moment of inertia about the center of mass. To put the angular momentum equation (9.20) in the more convenient body axes, we follow the same procedure as for the linear momentum equation above, except that $\mathbf{F}$ is now replaced by $\mathbf{M}$, and $m\mathbf{U}$ is replaced by $\mathbf{H}$.

$$\begin{aligned}
\bar{\bar{\mathbf{T}}}_b^e\,\mathbf{M}^b &= \frac{\mathrm{d}}{\mathrm{d}t}(\bar{\bar{\mathbf{T}}}_b^e\mathbf{H}^b) \\
&\;\;\vdots \\
&= \bar{\bar{\mathbf{T}}}_b^e\dot{\mathbf{H}}^b + (\boldsymbol{\Omega}\times\mathbf{H})^e
\end{aligned} \tag{9.22}$$

Premultiplying the final result by $\bar{\bar{\mathbf{T}}}_e^b$, and then replacing $\mathbf{H}^b$ by $\bar{\bar{\mathbf{I}}}^b\boldsymbol{\Omega}^b + \mathbf{h}^b$ gives the angular momentum equation for the aircraft in body axes.

$$\boxed{\mathbf{M}^b = \bar{\bar{\mathbf{I}}}^b\dot{\boldsymbol{\Omega}}^b + \boldsymbol{\Omega}^b\times\left(\bar{\bar{\mathbf{I}}}^b\boldsymbol{\Omega}^b + \mathbf{h}^b\right)} \tag{9.23}$$

9.8 Flight Dynamics Formulation

9.8.1 Variable and vector definitions

In this section we will adopt the following simplified notation for the aircraft motion parameters and aerodynamic forces and moments. This notation is fairly standard in the discipline of flight dynamics, stability, and control.

$$\mathbf{R}^e = \begin{Bmatrix} x^e \\ y^e \\ z^e \end{Bmatrix} \qquad \mathbf{U}^b = \begin{Bmatrix} u \\ v \\ w \end{Bmatrix} \qquad \mathbf{\Omega}^b = \begin{Bmatrix} p \\ q \\ r \end{Bmatrix} \qquad \mathbf{F}^b = \begin{Bmatrix} X \\ Y \\ Z \end{Bmatrix} \qquad \mathbf{M}^b = \begin{Bmatrix} \mathcal{L} \\ \mathcal{M} \\ \mathcal{N} \end{Bmatrix}$$

The components of the first three vectors above, together with the three Euler angles, are grouped in the following *state vector* with 12 components:

$$\mathbf{x}(t) = \{ \; x^e \; y^e \; z^e \quad \phi \; \theta \; \psi \quad u \; v \; w \quad p \; q \; r \; \}^T \tag{9.24}$$

As treated in Section 6.3, the aerodynamic forces X, Y, Z and moments $\mathcal{L}, \mathcal{M}, \mathcal{N}$ are functions of this state vector, and also of the *control vector* $\boldsymbol{\delta}$, which for a typical aircraft consists of aerodynamic control surface deflections and engine forces.

$$\boldsymbol{\delta}(t) = \{ \; \delta_a \; \delta_e \; \delta_r \quad \delta_f \; \delta_T \; \}^T \tag{9.25}$$

The first four components correspond to aileron, elevator, rudder, and flap deflections, and the last component is an engine-thrust variable. Unconventional aircraft may have other types of control variables.

9.8.2 General equations of motion

The 12 state vector components (9.24) are governed by 12 ODEs in time. These are the six kinematic equations for the aircraft position rate (9.11) and the Euler angle rates (9.16), and the six dynamic equations for the linear momentum rate (9.19) and the angular momentum rate (9.19). Using the new simplified notation defined above, these 12 equations are written out fully as follows.

$$\begin{aligned}
\dot{x}^e &= (\cos\theta\,\cos\psi)\,u + (\sin\phi\,\sin\theta\,\cos\psi - \cos\phi\,\sin\psi)\,v + (\cos\phi\,\sin\theta\,\cos\psi + \sin\phi\,\sin\psi)\,w \\
\dot{y}^e &= (\cos\theta\,\sin\psi)\,u + (\sin\phi\,\sin\theta\,\sin\psi + \cos\phi\,\cos\psi)\,v + (\cos\phi\,\sin\theta\,\sin\psi - \sin\phi\,\cos\psi)\,w \\
\dot{z}^e &= (-\sin\theta)\,u + (\sin\phi\,\cos\theta)\,v + (\cos\phi\,\cos\theta)\,w
\end{aligned} \tag{9.26}$$

$$\begin{aligned}
\dot{\phi} &= p + q\,\sin\phi\,\tan\theta + r\,\cos\phi\,\tan\theta \\
\dot{\theta} &= q\,\cos\phi - r\,\sin\phi \\
\dot{\psi} &= q\,\sin\phi/\cos\theta + r\,\cos\phi/\cos\theta
\end{aligned} \tag{9.27}$$

$$\begin{aligned}
X - mg\,\sin\theta &= m\,(\dot{u} + qw - rv) \\
Y + mg\,\sin\phi\,\cos\theta &= m\,(\dot{v} + ru - pw) \\
Z + mg\,\cos\phi\,\cos\theta &= m\,(\dot{w} + pv - qu)
\end{aligned} \tag{9.28}$$

$$\begin{aligned}
\mathcal{L} &= I_{xx}\,\dot{p} + I_{xy}\,\dot{q} + I_{xz}\,\dot{r} + (I_{zz} - I_{yy})\,qr + I_{yz}\,(q^2 - r^2) + I_{xz}\,pq - I_{xy}\,pr + h_z\,q - h_y\,r \\
\mathcal{M} &= I_{xy}\,\dot{p} + I_{yy}\,\dot{q} + I_{yz}\,\dot{r} + (I_{xx} - I_{zz})\,rp + I_{xz}\,(r^2 - p^2) + I_{xy}\,qr - I_{yz}\,qp + h_x\,r - h_z\,p \\
\mathcal{N} &= I_{xz}\,\dot{p} + I_{yz}\,\dot{q} + I_{zz}\,\dot{r} + (I_{zz} - I_{xx})\,pq + I_{xy}\,(p^2 - q^2) + I_{yz}\,rp - I_{xz}\,rq + h_y\,p - h_x\,q
\end{aligned} \tag{9.29}$$

When the three linear momentum equations (9.28) are multiplied by $1/m$, and the three angular momentum equations (9.29) are multiplied by the inverse of the moment of inertia tensor, $\bar{\bar{\mathbf{I}}}^{-1}$, they become explicit expressions for the linear and angular velocity rates, $\dot{u} = \ldots$, $\dot{p} = \ldots$, etc. All the 12 equations (9.26)–(9.29) then collectively have the classical state-space evolution form with 12 equation components:

$$\boxed{\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \boldsymbol{\delta})} \tag{9.30}$$

9.8.3 Linearized equations of motion

The state-space equation system (9.30) above is nonlinear. To make stability and control problems tractable, these equations are first put into *linearized* forms. We assume some steady flight in the *trim state* $\mathbf{x}_0$, $\boldsymbol{\delta}_0$. The trim state has $\dot{\mathbf{x}}_0 = \mathbf{0}$, except for the position rate and heading rate components.

$$\dot{x}^e = u_0^e \quad , \qquad \dot{y}^e = v_0^e \quad , \qquad \dot{z}^e = w_0^e \quad , \qquad \dot{\psi} = r_0^e$$

Straight flight has a zero heading rate $r_0^e = 0$, while steady turning flight has $r_0^e \neq 0$.

In the system equation (9.30), consider small perturbations $\Delta\mathbf{x}(t)$ and $\Delta\boldsymbol{\delta}(t)$ about the trim state, as shown in Figure 9.3.

$$\mathbf{x}(t) \;=\; \mathbf{x}_0 \;+\; \Delta\mathbf{x}(t) \tag{9.31}$$

$$\boldsymbol{\delta}(t) \;=\; \boldsymbol{\delta}_0 \;+\; \Delta\boldsymbol{\delta}(t) \tag{9.32}$$

$$\dot{\mathbf{x}}_0 \;+\; \Delta\dot{\mathbf{x}} \;=\; \mathbf{f}(\mathbf{x}_0+\Delta\mathbf{x}\,,\,\boldsymbol{\delta}_0+\Delta\boldsymbol{\delta}) \;\simeq\; \mathbf{f}(\mathbf{x}_0\,,\,\boldsymbol{\delta}_0) \;+\; \left.\frac{\mathrm{d}\mathbf{f}}{\mathrm{d}\mathbf{x}}\right|_{\mathbf{x}_0,\boldsymbol{\delta}_0} \Delta\mathbf{x} \;+\; \left.\frac{\mathrm{d}\mathbf{f}}{\mathrm{d}\boldsymbol{\delta}}\right|_{\mathbf{x}_0,\boldsymbol{\delta}_0} \Delta\boldsymbol{\delta} \tag{9.33}$$

Since the trim state is physical, it must by itself obey the equations of motion.

$$\dot{\mathbf{x}}_0 \;=\; \mathbf{f}(\mathbf{x}_0\,,\,\boldsymbol{\delta}_0) \tag{9.34}$$

Subtracting (9.34) from (9.33) gives the *linearized equations of motion* which govern the small state vector perturbations,

$$\Delta\dot{\mathbf{x}} \;=\; \bar{\bar{\mathbf{A}}}\,\Delta\mathbf{x} \;+\; \bar{\bar{\mathbf{B}}}\,\Delta\boldsymbol{\delta} \tag{9.35}$$

$$\bar{\bar{\mathbf{A}}} \;\equiv\; \left.\frac{\mathrm{d}\mathbf{f}}{\mathrm{d}\mathbf{x}}\right|_{\mathbf{x}_0,\boldsymbol{\delta}_0} \quad , \qquad \bar{\bar{\mathbf{B}}} \;\equiv\; \left.\frac{\mathrm{d}\mathbf{f}}{\mathrm{d}\boldsymbol{\delta}}\right|_{\mathbf{x}_0,\boldsymbol{\delta}_0}$$

with $\bar{\bar{\mathbf{A}}}$ and $\bar{\bar{\mathbf{B}}}$ being the *system Jacobian matrices* which depend on the trim state.

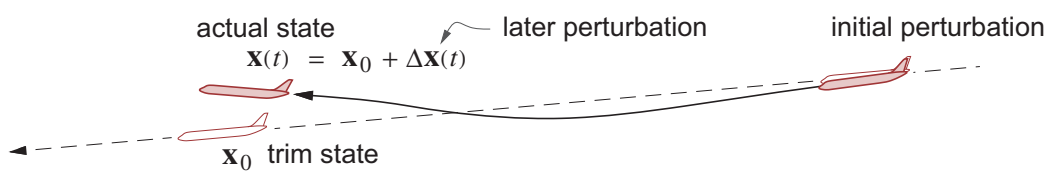

Figure 9.3: Aircraft state considered as a small perturbation from a trim state. Instability is indicated if the perturbation grows exponentially from an initial perturbation.

9.8.4 Natural response

For the *fixed-stick* case of no control inputs, $\Delta\boldsymbol{\delta} = \mathbf{0}$, the general solution of the Linear Time Invariant (LTI) ODE system (9.35) is a superposition of eight *flight-dynamics eigenmodes*,

$$\Delta\mathbf{x}(t) \;=\; \sum_{k=1}^{8} \mathbf{v}_k \,\exp(\lambda_k t) \quad , \qquad \lambda_k \;=\; \sigma_k + \mathrm{i}\,\omega_k \tag{9.36}$$

where $\lambda_{1\ldots8}$ are the nonzero eigenvalues and $\mathbf{v}_{1\ldots8}$ are the corresponding eigenvectors of the Jacobian matrix $\bar{\bar{\mathbf{A}}}$. Each of the remaining four zero eigenvalues corresponds to a shift in x, y, z, ψ, which has no influence on the dynamics and hence is excluded from the mode summation.

The magnitude of each ω_k and the sign and magnitude of σ_k indicate the nature of that mode. If $\omega_k = 0$ then the motion is monotonic, and if $\omega_k \neq 0$ then the motion is oscillatory. The latter actually consists of two complex-pair modes, $\lambda = \sigma \pm \mathrm{i}\,\omega$. If $\sigma_k > 0$ then that mode is *unstable*, meaning that the aircraft will exhibit exponential divergence from the trimmed flight state.

9.8.5 Symmetry

In the completely general case, the $\bar{\bar{\mathbf{A}}}$ and $\bar{\bar{\mathbf{B}}}$ matrices of the linearized dynamics system (9.35) are nearly full. But if we have

- geometric symmetry (the airplane is left/right symmetric), and
- aerodynamic symmetry with $v_0 = p_0 = r_0 = \phi_0 = 0$, and
- negligible onboard angular momentum, $\mathbf{h} \simeq \mathbf{0}$,

then $\bar{\bar{\mathbf{A}}}$ and $\bar{\bar{\mathbf{B}}}$ have the following form after the state vector and the equations are re-ordered as indicated. Zero or nearly zero matrix elements are shown blank.

$$\frac{\mathrm{d}}{\mathrm{d}t}\left\{\begin{array}{c}\Delta u\\ \Delta w\\ \Delta q\\ \Delta\theta\\ \hline \Delta v\\ \Delta p\\ \Delta r\\ \Delta\phi\\ \hline \Delta x^e\\ \Delta y^e\\ \Delta z^e\\ \Delta\psi\end{array}\right\} = \left[\begin{array}{cccc|cccc|c}\cdot&\cdot&\cdot&\cdot&&&&&\\ \cdot&\cdot&\cdot&\cdot&&&&&\\ \cdot&\cdot&\cdot&\cdot&&&&&\\ \cdot&\cdot&\cdot&\cdot&&&&&\\ \hline &&&&\cdot&\cdot&\cdot&\cdot&\\ &&&&\cdot&\cdot&\cdot&\cdot&\\ &&&&\cdot&\cdot&\cdot&\cdot&\\ &&&&\cdot&\cdot&\cdot&\cdot&\\ \hline \cdot&\cdot&\cdot&\cdot&\cdot&\cdot&\cdot&\cdot&\\ \cdot&\cdot&\cdot&\cdot&\cdot&\cdot&\cdot&\cdot&\\ \cdot&\cdot&\cdot&\cdot&\cdot&\cdot&\cdot&\cdot&\\ \cdot&\cdot&\cdot&\cdot&\cdot&\cdot&\cdot&\cdot&\end{array}\right]\left\{\begin{array}{c}\Delta u\\ \Delta w\\ \Delta q\\ \Delta\theta\\ \hline \Delta v\\ \Delta p\\ \Delta r\\ \Delta\phi\\ \hline \Delta x^e\\ \Delta y^e\\ \Delta z^e\\ \Delta\psi\end{array}\right\} + \left[\begin{array}{ccc|cc}\cdot&\cdot&\cdot&&\\ \cdot&\cdot&\cdot&&\\ \cdot&\cdot&\cdot&&\\ \hline &&&\cdot&\cdot\\ &&&\cdot&\cdot\\ &&&\cdot&\cdot\\ \hline &&&&\\ &&&&\\ &&&&\\ &&&&\end{array}\right]\left\{\begin{array}{c}\Delta\delta_T\\ \Delta\delta_f\\ \Delta\delta_e\\ \hline \Delta\delta_a\\ \Delta\delta_r\end{array}\right\} \tag{9.37}$$

The first four rows constitute the *Longitudinal Dynamics* subset, the middle four rows are the *Lateral Dynamics* subset, and the last four rows are the *Navigation* subset.

The last four navigation variables $\Delta x^e, \Delta y^e, \Delta z^e, \Delta\psi$ generally have zero matrix columns, even for asymmetric aircraft. This implies that the matrix will have at least four zero eigenvalues, $\lambda_{9,10,11,12} = 0$, whose corresponding eigenmodes contain only the steady displacements of the four navigation variables.

$$\begin{array}{rcl}\mathbf{v}_9\ \exp(0t) &=& \{\quad\quad\quad|\quad\quad\quad|\Delta x^e\quad\quad\}^T\\ \mathbf{v}_{10}\ \exp(0t) &=& \{\quad\quad\quad|\quad\quad\quad|\quad\Delta y^e\quad\}^T\\ \mathbf{v}_{11}\ \exp(0t) &=& \{\quad\quad\quad|\quad\quad\quad|\quad\quad\Delta z^e\quad\}^T\\ \mathbf{v}_{12}\ \exp(0t) &=& \{\quad\quad\quad|\quad\quad\quad|\quad\quad\quad\Delta\psi\}^T\end{array}$$

This is expected from coordinate invariance – the dynamics cannot be affected if the airplane position is displaced, or if it is pointed in another direction in the plane parallel to the Earth surface. This justifies excluding these four trivial steady displacement modes from the dynamical mode summation (9.36).

One notable exception occurs if atmospheric property variation is significant over the vertical extent of the motion. In this case there will be some dependence on the Δz^e variable, and there will be only three trivial modes present in the system.

9.9 Aerodynamic Force and Moment Linearizations

Force and moment coefficients have been already defined in Section 6.3. In the most common notation used in flight dynamics these are

$$\begin{array}{rclrcl}C_x &\equiv& X/QS & C_\ell &\equiv& \mathcal{L}/QSb\\ C_y &\equiv& Y/QS & C_m &\equiv& \mathcal{M}/QSc\\ C_z &\equiv& Z/QS & C_n &\equiv& \mathcal{N}/QSb\end{array} \tag{9.38}$$

where S, b, c are the reference area, span, chord. To avoid conflict with the pitch rate q, the symbol Q is used to denote the reference freestream dynamic pressure in lieu of the alternative q_∞.

$$Q \;\equiv\; \tfrac{1}{2}\rho V^2 \;=\; \tfrac{1}{2}\rho(u^2{+}v^2{+}w^2) \tag{9.39}$$

Perturbations of velocity, flow angles, and normalized rates about the trim state are related to the state variable perturbations as follows.

$$\begin{aligned}
\Delta\bar{u} &\equiv \frac{\Delta u}{u_0} & \Delta\bar{p} &\equiv \Delta\left(\frac{pb}{2V}\right) \simeq \Delta p\,\frac{b}{2u_0}\\
\Delta\beta &\equiv \Delta\left(\arctan\frac{v}{\sqrt{u^2{+}w^2}}\right) \simeq \frac{\Delta v}{u_0} & \Delta\bar{q} &\equiv \Delta\left(\frac{qc}{2V}\right) \simeq \Delta q\,\frac{c}{2u_0}\\
\Delta\alpha &\equiv \Delta\left(\arctan\frac{w}{u}\right) \simeq \frac{\Delta w}{u_0} & \Delta\bar{r} &\equiv \Delta\left(\frac{rb}{2V}\right) \simeq \Delta r\,\frac{b}{2u_0}
\end{aligned} \tag{9.40}$$

The dynamic pressure perturbation is linearized in the same manner.

$$\Delta Q \;=\; \Delta\left[\tfrac{1}{2}\rho(u^2+v^2+w^2)\right] \;\simeq\; \rho\,(u_0\,\Delta u \,+\, v_0\,\Delta v \,+\, w_0\,\Delta w) \;\simeq\; \rho u_0\,\Delta u \;=\; 2Q\,\frac{\Delta u}{u_0} \tag{9.41}$$

The coefficient perturbations give the state variable and control variable perturbations via the stability and control derivatives as follows. Again, blank matrix elements indicate derivatives which are zero for a symmetric aircraft in a symmetric trim state flight condition. An asymmetric trim state would have all the elements nonzero.

$$\begin{Bmatrix}\Delta C_x\\ \Delta C_y\\ \Delta C_z\end{Bmatrix} = \begin{bmatrix} C_{x_u} & & C_{x_\alpha}\\ & C_{y_\beta} & \\ C_{z_u} & & C_{z_\alpha}\end{bmatrix}\begin{Bmatrix}\Delta\bar{u}\\ \Delta\beta\\ \Delta\alpha\end{Bmatrix} + \begin{bmatrix} & C_{x_q} & \\ C_{y_p} & & C_{y_r}\\ & C_{z_q} & \end{bmatrix}\begin{Bmatrix}\Delta\bar{p}\\ \Delta\bar{q}\\ \Delta\bar{r}\end{Bmatrix} + \begin{bmatrix} C_{x_{\delta_T}} & & \\ & & \\ & & C_{z_{\delta_f}}\end{bmatrix}\begin{Bmatrix}\Delta\delta_T\\ \Delta\delta_f\end{Bmatrix} \tag{9.42}$$

$$\begin{Bmatrix}\Delta C_\ell\\ \Delta C_m\\ \Delta C_n\end{Bmatrix} = \begin{bmatrix} & C_{\ell_\beta} & \\ C_{m_u} & & C_{m_\alpha}\\ & C_{n_\beta} & \end{bmatrix}\begin{Bmatrix}\Delta\bar{u}\\ \Delta\beta\\ \Delta\alpha\end{Bmatrix} + \begin{bmatrix} C_{\ell_p} & & C_{\ell_r}\\ & C_{m_q} & \\ C_{n_p} & & C_{n_r}\end{bmatrix}\begin{Bmatrix}\Delta\bar{p}\\ \Delta\bar{q}\\ \Delta\bar{r}\end{Bmatrix} + \begin{bmatrix} C_{\ell_{\delta_a}} & & \\ & C_{m_{\delta_e}} & \\ & & C_{n_{\delta_r}}\end{bmatrix}\begin{Bmatrix}\Delta\delta_a\\ \Delta\delta_e\\ \Delta\delta_r\end{Bmatrix} \tag{9.43}$$

The dimensional force and moment perturbations about the trim state are now expressed in terms of the coefficient and dynamic pressure perturbations obtained above.

$$\begin{aligned}
\Delta X &= \Delta\,(QS\,C_x) \simeq QS\left(2C_{x_0}\,\Delta\bar{u} \,+\, C_{x_\alpha}\Delta\alpha \,+\, C_{x_q}\Delta\bar{q} \,+\, C_{x_{\delta_T}}\Delta\delta_T\right)\\
\Delta Y &= \Delta\,(QS\,C_y) \simeq QS\left(C_{y_\beta}\Delta\beta \,+\, C_{y_p}\Delta\bar{p} \,+\, C_{y_r}\Delta\bar{r}\right)\\
\Delta Z &= \Delta\,(QS\,C_z) \simeq QS\left(2C_{z_0}\,\Delta\bar{u} \,+\, C_{z_\alpha}\Delta\alpha \,+\, C_{z_q}\Delta\bar{q} \,+\, C_{z_{\delta_f}}\Delta\delta_f \,+\, C_{z_{\delta_e}}\Delta\delta_e\right)
\end{aligned} \tag{9.44}$$

$$\begin{aligned}
\Delta\mathcal{L} &= \Delta\,(QSb\,C_\ell) \simeq QSb\left(C_{\ell_\beta}\Delta\beta \,+\, C_{\ell_p}\Delta\bar{p} \,+\, C_{\ell_r}\Delta\bar{r} \,+\, C_{\ell_{\delta_a}}\Delta\delta_a\right)\\
\Delta\mathcal{M} &= \Delta\,(QSc\,C_m) \simeq QSc\left(2C_{m_0}\,\Delta\bar{u} \,+\, C_{m_\alpha}\Delta\alpha \,+\, C_{m_q}\Delta\bar{q} \,+\, C_{m_{\delta_e}}\Delta\delta_e\right)\\
\Delta\mathcal{N} &= \Delta\,(QSb\,C_n) \simeq QSb\left(C_{n_\beta}\Delta\beta \,+\, C_{n_p}\Delta\bar{p} \,+\, C_{n_r}\Delta\bar{r} \,+\, C_{n_{\delta_r}}\Delta\delta_r\right)
\end{aligned} \tag{9.45}$$

As discussed in Section 6.3.2, aero-lag effects also make the forces and moments depend on the angle of attack rate. This could be represented above by additional terms with the associated stability derivatives, such as $C_{z_{\dot\alpha}}\Delta\dot\alpha$, $C_{m_{\dot\alpha}}\Delta\dot\alpha$, etc. These are left out here for brevity.

9.10 Stability Derivative Specification

Aerodynamic force and moment coefficients and their derivatives are usually specified in the *stability axes* $(x^{b'}, z^{b'})$, which are rotated about $y^b = y^{b'}$ by the freestream angle of attack α, as shown in Figure 9.4. The main reason for this axis choice is that the stability derivatives in stability axes exhibit a minimal variation with the trim-state α_0 within the normal operating range. Specifically, in unstalled flight, the lateral cross-derivatives C'_{ℓ_r} and C'_{n_p} are very nearly proportional to the trim C_{L_0}, and the remaining derivatives are nearly the same for all C_{L_0} values. This considerably simplifies the stability and control specification of an aircraft for all trim conditions. The 2×2 rotation matrix $\bar{\bar{\mathbf{S}}}$ then defines the body-axis xz vector components in terms of the $x'z'$ components, shown in Figure 9.4. This is effectively the same as matrix $\bar{\bar{\mathbf{T}}}^{\mathrm{s}}$ defined by (6.4) in Section 6.2.1.

$$\bar{\bar{\mathbf{S}}}(\alpha) \equiv \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix} \quad , \quad \frac{\mathrm{d}\bar{\bar{\mathbf{S}}}}{\mathrm{d}\alpha}(\alpha) = \begin{bmatrix} -\sin\alpha & -\cos\alpha \\ \cos\alpha & -\sin\alpha \end{bmatrix} \tag{9.46}$$

$$\begin{Bmatrix} C_x \\ C_z \end{Bmatrix} = \begin{bmatrix} & \\ & \bar{\bar{\mathbf{S}}} & \\ & & \end{bmatrix} \begin{Bmatrix} -C_D \\ -C_L \end{Bmatrix} \quad , \quad \begin{Bmatrix} C_\ell \\ C_n \end{Bmatrix} = \begin{bmatrix} & \\ & \bar{\bar{\mathbf{S}}} & \\ & & \end{bmatrix} \begin{Bmatrix} C'_\ell \\ C'_n \end{Bmatrix} \quad , \quad \begin{Bmatrix} \bar{p} \\ \bar{r} \end{Bmatrix} = \begin{bmatrix} & \\ & \bar{\bar{\mathbf{S}}} & \\ & & \end{bmatrix} \begin{Bmatrix} \bar{p}' \\ \bar{r}' \end{Bmatrix} \tag{9.47}$$

The remaining y-axis components $C_y, C_m, \bar{q}$ are the same between the two axis systems.

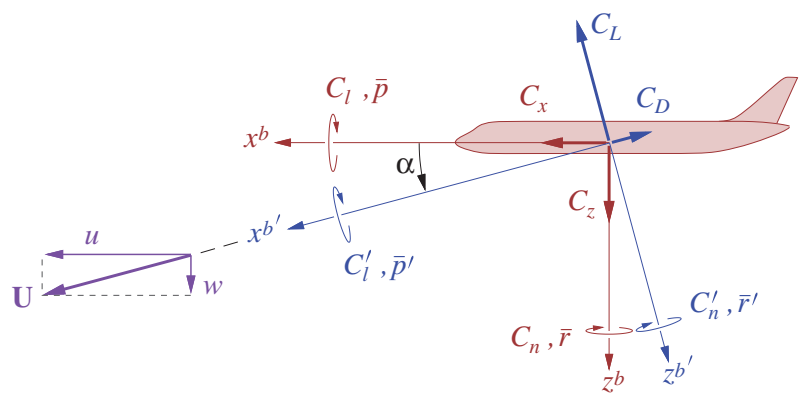

Figure 9.4: Force and moment coefficients in body and stability axes. This is the non-dimensional equivalent to Figure 6.2.

The body-axis derivatives with respect to α, all defined at the trim state, can now be computed:

$$\begin{Bmatrix} C_{x_\alpha} \\ C_{z_\alpha} \end{Bmatrix} = \begin{bmatrix} & \\ & \left.\frac{\mathrm{d}\bar{\bar{\mathbf{S}}}}{\mathrm{d}\alpha}\right|_0 & \\ & & \end{bmatrix} \begin{Bmatrix} -C_{D_0} \\ -C_{L_0} \end{Bmatrix} + \begin{bmatrix} & \\ & \bar{\bar{\mathbf{S}}}_0 & \\ & & \end{bmatrix} \begin{Bmatrix} -C_{D_\alpha} \\ -C_{L_\alpha} \end{Bmatrix} \simeq \begin{Bmatrix} C_{L_0} - C_{D_\alpha} \\ -C_{L_\alpha} \end{Bmatrix} \tag{9.48}$$

The body-axis moment derivatives with respect to β and control deflections are similarly computed.

$$\begin{Bmatrix} C'_{\ell_\beta} \\ C'_{n_\beta} \end{Bmatrix} = \begin{bmatrix} & \\ & \bar{\bar{\mathbf{S}}}_0 & \\ & & \end{bmatrix} \begin{Bmatrix} C'_{\ell_\beta} \\ C'_{n_\beta} \end{Bmatrix} \simeq \begin{Bmatrix} C'_{\ell_\beta} - \alpha_0\, C'_{n_\beta} \\ C'_{n_\beta} + \alpha_0\, C'_{\ell_\beta} \end{Bmatrix} \tag{9.49}$$

$$\begin{Bmatrix} C'_{\ell_\delta} \\ C'_{n_\delta} \end{Bmatrix} = \begin{bmatrix} & \\ & \bar{\bar{\mathbf{S}}}_0 & \\ & & \end{bmatrix} \begin{Bmatrix} C'_{\ell_\delta} \\ C'_{n_\delta} \end{Bmatrix} \simeq \begin{Bmatrix} C'_{\ell_\delta} - \alpha_0\, C'_{n_\delta} \\ C'_{n_\delta} + \alpha_0\, C'_{\ell_\delta} \end{Bmatrix} \tag{9.50}$$

The sideforce derivatives with respect to the body-axis rates are obtained with the reverse rotation by the transpose of $\bar{\bar{\mathbf{S}}}$:

$$\begin{pmatrix} C_{y_p} & C_{y_r} \end{pmatrix} = \begin{pmatrix} C'_{y_p} & C'_{y_r} \end{pmatrix} \begin{bmatrix} & \\ & \bar{\bar{\mathbf{S}}}_0^T & \\ & & \end{bmatrix} \simeq \begin{pmatrix} C'_{y_p} - \alpha_0\, C'_{y_r} & C'_{y_r} + \alpha_0\, C'_{y_p} \end{pmatrix} \tag{9.51}$$

Converting the moment derivatives with respect to $\bar{p}, \bar{r}$ into the body axes requires two rotations, a pre-rotation by $\bar{\bar{\mathbf{S}}}$ for the moments and a reverse post-rotation by $\bar{\bar{\mathbf{S}}}^T$ for the rates.

$$\begin{bmatrix} C_{\ell_p} & C_{\ell_r} \\ C_{n_p} & C_{n_r} \end{bmatrix} = \begin{bmatrix} & \\ & \bar{\bar{\mathbf{S}}}_0 & \\ & \end{bmatrix} \begin{bmatrix} C'_{\ell_p} & C'_{\ell_r} \\ C'_{n_p} & C'_{n_r} \end{bmatrix} \begin{bmatrix} & \\ & \bar{\bar{\mathbf{S}}}_0^T & \\ & \end{bmatrix} \tag{9.52}$$

9.11 Longitudinal Dynamics Subset

The longitudinal-dynamics system is the upper 4×4 part of the overall system (9.37).

$$\begin{Bmatrix} \Delta\dot{u} \\ \Delta\dot{w} \\ \Delta\dot{q} \\ \Delta\dot{\theta} \end{Bmatrix} = \begin{bmatrix} X_u & X_w & 0 & -g \\ Z_u & Z_w & u_0 & 0 \\ M_u & M_w & M_q & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{Bmatrix} \Delta u \\ \Delta w \\ \Delta q \\ \Delta\theta \end{Bmatrix} + \begin{bmatrix} X_{\delta_T} & 0 & 0 \\ 0 & Z_{\delta_f} & 0 \\ 0 & 0 & M_{\delta_e} \\ 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} \Delta\delta_T \\ \Delta\delta_f \\ \Delta\delta_e \end{Bmatrix} \tag{9.53}$$

The matrix elements above are the following dimensional stability derivatives.

$$\begin{aligned}
X_u &= \frac{QS}{m}\frac{1}{u_0} C_{x_u} & Z_u &= \frac{QS}{m}\frac{1}{u_0} C_{z_u} & M_u &= \frac{QSc}{I_y}\frac{1}{u_0} C_{m_u} \\
X_w &= \frac{QS}{m}\frac{1}{u_0} C_{x_\alpha} & Z_w &= \frac{QS}{m}\frac{1}{u_0} C_{z_\alpha} & M_w &= \frac{QSc}{I_y}\frac{1}{u_0} C_{m_\alpha} \\
X_q &= \frac{QS}{m}\frac{c}{2u_0} C_{x_q} & Z_q &= \frac{QS}{m}\frac{c}{2u_0} C_{z_q} & M_q &= \frac{QSc}{I_y}\frac{c}{2u_0} C_{m_q} \\
X_{\delta_T} &= \frac{QS}{m} C_{x_{\delta_T}} & Z_{\delta_f} &= \frac{QS}{m} C_{z_{\delta_f}} & M_{\delta_e} &= \frac{QSc}{I_y} C_{m_{\delta_e}}
\end{aligned} \tag{9.54}$$

These assume that the moment of inertia tensor $\bar{\bar{\mathbf{I}}}$ has only the single pitch inertia element $I_{yy} \equiv I_y$, with I_{xy} and I_{yz} being zero, which is consistent with the assumed left/right symmetry.

For a typical pitch-stable aircraft in fixed-stick flight with $\Delta\delta_T = \Delta\delta_f = \Delta\delta_e = 0$, the four eigenvalues of the longitudinal system (9.53) are shown in Figure 9.5, and have the following characteristics.

$\sigma_{\rm ph} \pm i\omega_{\rm ph}$	Phugoid mode	(slow, weakly damped)
$\sigma_{\rm SP} \pm i\omega_{\rm SP}$	Short-Period mode	(fast, well-damped)

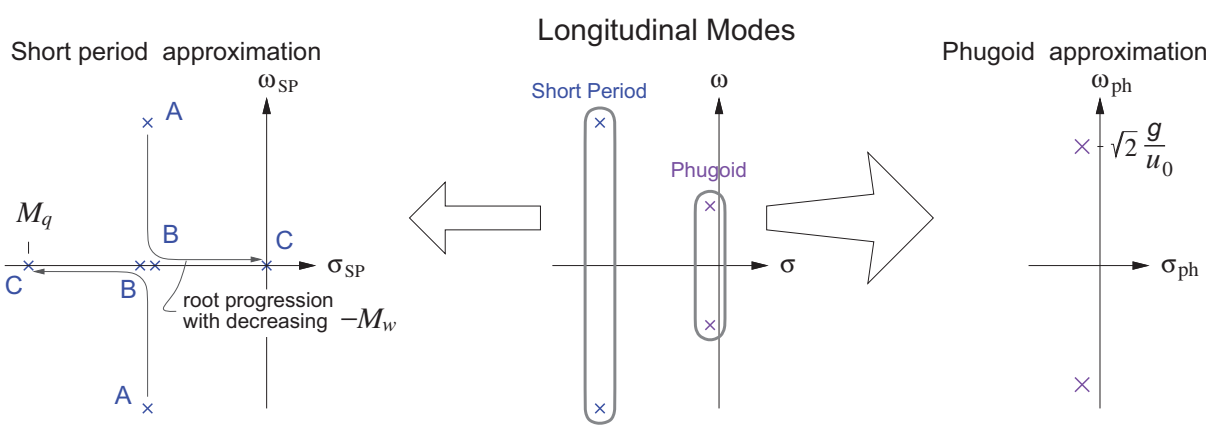

Figure 9.5: Typical longitudinal dynamics eigenvalues of a conventional aircraft. Short period and phugoid eigenvalues closely match those given by their respective approximation models.

These two eigenvalue pairs have good *spectral separation*, so that they can be estimated reasonably well in isolation by assuming that some motion components are negligible. This will give some insight into the key aircraft aerodynamic parameters which influence these eigenmodes. The assumed motions are shown in Figures 9.6 and 9.7, and are analyzed in the following sections.

9.11.1 Phugoid approximation

Here we assume pitch equilibrium, or $\Delta\mathcal{M}=0$, which implies a negligible pitch rate and pitch acceleration, $\Delta q \simeq 0$, $\Delta\dot{q} \simeq 0$. The airplane is thus assumed to remain in pitch trim, implying a fixed-α motion so that $\Delta w = u_0\,\Delta\alpha = 0$, as shown in Figure 9.6.

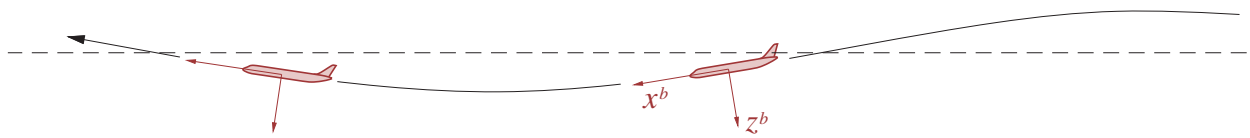

Figure 9.6: Assumed motion for the phugoid approximation.

The full longitudinal system (9.53) then simplifies to the "phugoid" system:

$$\begin{Bmatrix} \Delta\dot{u} \\ \Delta\dot{\theta} \end{Bmatrix} = \begin{bmatrix} X_u & -g \\ -Z_u/u_0 & 0 \end{bmatrix} \begin{Bmatrix} \Delta u \\ \Delta\theta \end{Bmatrix} + \begin{bmatrix} X_{\delta_T} \\ 0 \end{bmatrix} \{\Delta\delta_T\} \tag{9.55}$$

Its 2×2 matrix has the two phugoid eigenvalues

$$\lambda_{1,2} \equiv \sigma_{\rm ph} \pm {\rm i}\,\omega_{\rm ph} = \frac{1}{2}\left[X_u \pm \sqrt{X_u^2 + 4gZ_u/u_0}\,\right]$$

which have the following phugoid frequency and damping ratio.

$$\omega_{\rm ph} = \frac{1}{2}\sqrt{-X_u^2 - 4gZ_u/u_0} \simeq \sqrt{2}\,\frac{g}{u_0} \qquad \text{(slow)} \tag{9.56}$$

$$\zeta_{\rm ph} \equiv \frac{-\sigma_{\rm ph}}{\sqrt{\sigma_{\rm ph}^2 + \omega_{\rm ph}^2}} \simeq \frac{1}{\sqrt{2}}\,\frac{C_{D_0}}{C_{L_0}} \qquad \text{(weakly damped)} \tag{9.57}$$

The corresponding eigenmodes in the sum (9.36) then give the natural response of the aircraft to pitch perturbations in the uncontrolled or "fixed-stick" case with $\Delta\delta_T = 0$.

For high speed aircraft the phugoid period $2\pi/\omega_{\rm ph}$ is measured in minutes, which is a rather slow motion compared to more typical aircraft maneuvers. For clean airplanes with large lift/drag ratios $L/D = C_{L_0}/C_{D_0}$, the phugoid damping ratio is seen to be especially small. On such aircraft, adequate suppression of unwanted phugoid motions requires either an autopilot or active pitch control by the pilot.

9.11.2 Short-period approximation

Here we assume that the pitching motions are too fast for the airplane to respond significantly to the resulting lift changes, so that during the pitch oscillations it travels in a straight line at a fixed speed. The straight-line motion implies $\Delta u = 0$ and $\Delta w = u_0\,\Delta\theta$, as shown in Figure 9.7.

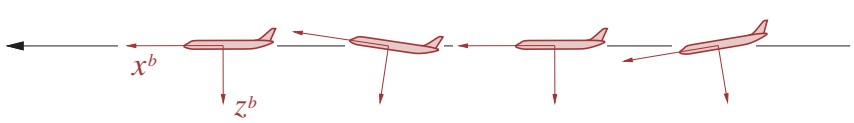

Figure 9.7: Assumed motion for the short-period approximation.

This reduces the full longitudinal system (9.53) to the smaller 2×2 "short period" system:

$$\begin{Bmatrix} \Delta\dot{q} \\ \Delta\dot{\theta} \end{Bmatrix} = \begin{bmatrix} M_q & u_0 M_w \\ 1 & 0 \end{bmatrix} \begin{Bmatrix} \Delta q \\ \Delta\theta \end{Bmatrix} + \begin{bmatrix} M_{\delta_e} \\ 0 \end{bmatrix} \{\,\Delta\delta_e\,\} \tag{9.58}$$

For the fixed-stick case $\Delta\delta_e = 0$, this has the following two short-period eigenvalues.

$$\lambda_{1,2} \equiv \sigma_{\rm SP} \pm {\rm i}\,\omega_{\rm SP} = \frac{1}{2}\left[M_q \pm \sqrt{M_q^2 + 4u_0 M_w}\,\right]$$

These eigenvalues strongly depend on the pitch stability derivative $M_w \sim C_{m_\alpha}$, which is normally negative. As M_w increases (becomes less negative), there are three distinct limiting cases, labeled A,B,C in Figure 9.5.

$$\begin{array}{llll}
\text{A)}\ M_w < -M_q^2/4u_0: & \sigma_{\rm SP} = \frac{1}{2}M_q\ , & \omega_{\rm SP} = \frac{1}{2}\sqrt{-M_q^2 - 4u_0 M_w} & \text{(damped oscillatory)} \\
\text{B)}\ M_w = -M_q^2/4u_0: & \sigma_{\rm SP} = \frac{1}{2}M_q\ , & \frac{1}{2}M_q & \text{(critically damped)} \\
\text{C)}\ M_w = 0 \quad\quad : & \sigma_{\rm SP} = M_q\ , & 0 & \text{(instability threshold)}
\end{array} \tag{9.59}$$

The gradual increase of M_w, or equivalently of C_{m_α}, can in practice be caused by a rearward movement of the center of mass, or a gradual reduction in the horizontal tail area or moment arm. Case C indicates that instability will occur when C_{m_α} becomes positive.

In reality, case C invalidates the short-period approximations. These assumed that the short-period motion was fast compared to the phugoid's frequency, but case C shows a stationary short-period mode with a zero frequency and zero time constant. Hence, it is necessary to examine this case in more detail.

The actual behavior of the four eigenvalues of the full unsimplified longitudinal-dynamics system (9.53) is shown in the Figure 9.8. Note that it is actually one of the phugoid eigenvalues which first becomes unstable. The associated eigenmode cannot be categorized as either a short-period motion or a phugoid motion, but is instead a monotonic pitch divergence. Nevertheless, despite the weakness of the short-period approximation at the predicted onset of instability, the $C_{m_\alpha} \geq 0$ criterion obtained from this approximation predicts the instability of the full unsimplified longitudinal system quite well.

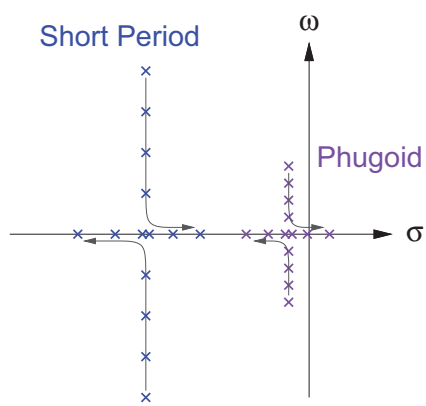

Figure 9.8: Progression of eigenvalues of complete longitudinal dynamics system as M_w (or C_{m_α}) becomes less negative and crosses zero.

9.12 Lateral Dynamics Subset

It is assumed that the aircraft has y-symmetry, so the products of inertia I_{xy} and I_{yz} vanish. Also, the xz body axes are assumed to be aligned with the inertia tensor's principal xz axes, so in these axes the I_{xz} product vanishes as well. With these assumptions the inertia tensor is then diagonal.

$$\bar{\bar{\mathbf{I}}} \equiv \begin{bmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{xy} & I_{yy} & I_{yz} \\ I_{xz} & I_{yz} & I_{zz} \end{bmatrix} = \begin{bmatrix} I_x & & \\ & I_y & \\ & & I_z \end{bmatrix}$$

The lateral-dynamics system, which is the middle 4×4 part of the overall system (9.37), then has the form

$$\begin{Bmatrix} \Delta\dot{v} \\ \Delta\dot{p} \\ \Delta\dot{r} \\ \Delta\dot{\phi} \end{Bmatrix} = \begin{bmatrix} Y_v & Y_p & Y_r - u_0 & g\cos\theta_0 \\ L_v & L_p & L_r & 0 \\ N_v & N_p & N_r & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{Bmatrix} \Delta v \\ \Delta p \\ \Delta r \\ \Delta\phi \end{Bmatrix} + \begin{bmatrix} Y_{\delta_a} & Y_{\delta_r} \\ L_{\delta_a} & L_{\delta_r} \\ N_{\delta_a} & N_{\delta_r} \\ 0 & 0 \end{bmatrix} \begin{Bmatrix} \Delta\delta_a \\ \Delta\delta_r \end{Bmatrix} \tag{9.60}$$

in which the matrix elements are the following dimensional stability derivatives.

$$\begin{aligned}
Y_v &= \frac{QS}{m}\frac{1}{u_0}C_{y_\beta} & L_v &= \frac{QSb}{I_x}\frac{1}{u_0}C_{\ell_\beta} & N_v &= \frac{QSb}{I_z}\frac{1}{u_0}C_{n_\beta} \\
Y_p &= \frac{QS}{m}\frac{b}{2u_0}C_{y_p} & L_p &= \frac{QSb}{I_x}\frac{b}{2u_0}C_{\ell_p} & N_p &= \frac{QSb}{I_z}\frac{b}{2u_0}C_{n_p} \\
Y_r &= \frac{QS}{m}\frac{b}{2u_0}C_{y_r} & L_r &= \frac{QSb}{I_x}\frac{b}{2u_0}C_{\ell_r} & N_r &= \frac{QSb}{I_z}\frac{b}{2u_0}C_{n_r} \\
Y_{\delta_a} &= \frac{QS}{m}C_{y_{\delta_a}} & L_{\delta_a} &= \frac{QSb}{I_x}C_{\ell_{\delta_a}} & N_{\delta_a} &= \frac{QSb}{I_z}C_{n_{\delta_a}} \\
Y_{\delta_r} &= \frac{QS}{m}C_{y_{\delta_r}} & L_{\delta_r} &= \frac{QSb}{I_x}C_{\ell_{\delta_r}} & N_{\delta_r} &= \frac{QSb}{I_z}C_{n_{\delta_r}}
\end{aligned} \tag{9.61}$$

For a typical aircraft in fixed-stick flight with $\Delta\delta_a = \Delta\delta_r = 0$, the four eigenvalues of the lateral system (9.60) are shown in Figure 9.9, and have the following characteristics.

σ_{roll}	Roll-subsidence mode	(very fast, damped)
σ_{spiral}	Spiral mode	(very slow, stable or unstable)
$\sigma_{\text{DR}} \pm \mathrm{i}\,\omega_{\text{DR}}$	Dutch Roll mode	(oscillatory, moderately damped)

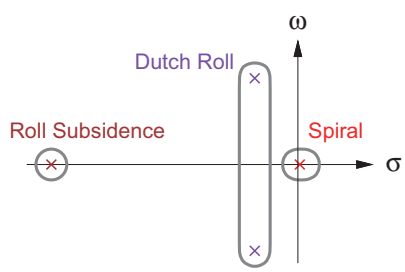

Figure 9.9: Typical lateral dynamics eigenvalues of a conventional aircraft.

As in the longitudinal-dynamics case, we will now obtain approximate expressions for these eigenvalues by assuming a simplified motion for the corresponding mode. The analysis follows that of Nelson [66].

9.12.1 Roll-subsidence approximation

For the roll-subsidence mode we assume a pure rolling motion about the x^b axis, shown in Figure 9.10.

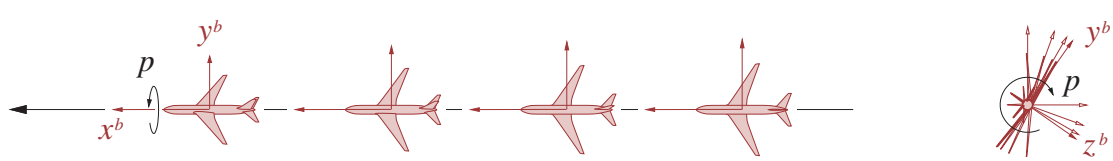

Figure 9.10: Assumed motion for the roll-subsidence approximation.

In this case only $\Delta\phi$ and $\Delta\dot{\phi} = \Delta p$ are nonzero, so the lateral-dynamics system (9.60) reduces to the following single first-order ODE.

$$\{\,\Delta\dot{p}\,\} \;=\; [\;L_p\;]\,\{\,\Delta p\,\} \;+\; [\;L_{\delta_a}\;\;L_{\delta_r}\,]\begin{Bmatrix}\Delta\delta_a\\ \Delta\delta_r\end{Bmatrix} \tag{9.62}$$

Its eigenvalue for the fixed-stick case $\Delta\delta_a = \Delta\delta_r = 0$ follows immediately.

$$\sigma_{\text{roll}} \;\simeq\; L_p \;=\; \frac{QSb^2}{2I_x u_0}\,C_{\ell_p} \tag{9.63}$$

Since C_{ℓ_p} is always negative, this corresponds to a monotonic decay in the roll rate. For most aircraft the characteristic damping time constant $1/\sigma_{\text{roll}}$ is less than one second.

9.12.2 Spiral approximation

For this mode we assume that the roll accelerations are negligible, and only yaw motions dominate as shown in Figure 9.11. Setting $\Delta\dot{p} = 0$ in the second roll-moment line in the lateral system (9.60) then gives

$$\begin{aligned} 0 \;&=\; L_v\Delta v \;+\; L_r\Delta r \\ \text{or}\qquad \Delta v \;&=\; -\frac{L_r}{L_v}\,\Delta r. \end{aligned} \tag{9.64}$$

Combining this with the fourth yaw-moment line gives

$$\{\,\Delta\dot{r}\,\} \;=\; \left[N_r - N_v\frac{L_r}{L_v}\right]\{\,\Delta r\,\} \tag{9.65}$$

whose eigenvalue is

$$\sigma_{\text{spiral}} \;\simeq\; N_r - N_v\frac{L_r}{L_v} \;=\; \frac{QSb^2}{2I_z u_0}\left(C_{n_r} - C_{n_\beta}\frac{C_{\ell_r}}{C_{\ell_\beta}}\right) \tag{9.66}$$

which corresponds to monotonic yaw motion. In reality some roll motion (into the turn) is also present, since roll and yaw motions are always coupled to some extent in the full lateral system (9.60).

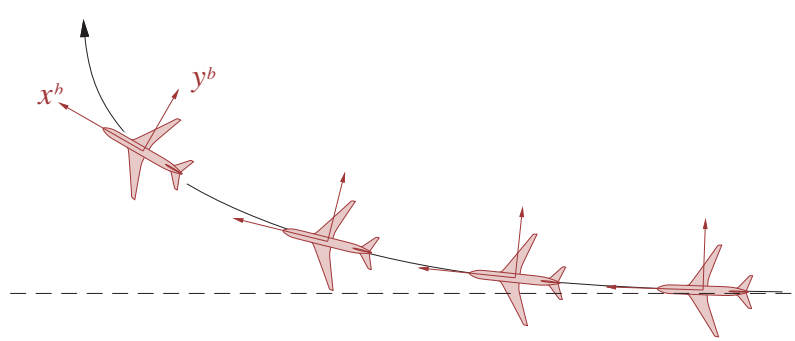

Figure 9.11: Assumed motion for the spiral-mode approximation. The tightening rate of the spiral path is shown greatly exaggerated.

The spiral mode can be either stable or unstable, depending on the values of the four stability derivatives appearing in (9.66). For conventional aircraft configurations, increasing the vertical-tail arm, increasing the dihedral, and decreasing the trim lift coefficient C_{L_0} all tend to make the spiral mode more stable. The magnitude of σ_{spiral} also scales roughly with the vertical tail area via the C_{n_r} and C_{n_β} derivatives

in (9.66). Therefore, increasing the vertical tail size will further stabilize a spirally-stable aircraft, and further destabilize a spirally-unstable aircraft.

Most aircraft are slightly spirally unstable, although the instability is slow enough to be easily controlled by a pilot who is attentive to the bank angle, either by observing the horizon or the artificial-horizon instrument. A wing-leveling autopilot can perform the same function without pilot input.

9.12.3 Dutch-roll approximation

The Dutch roll mode is an analog of the short-period mode, in that it mainly consists of relatively fast yawing motions, but which are now about the vertical axis. However, the Dutch roll mode is more complicated because it usually also involves significant roll motions. It is these secondary roll motions, which resemble those of a (Dutch) skater, which give the mode its name. Nevertheless, it is still useful to idealize the motion as shown in Figure 9.12, for which we have $\Delta v = -u_0\,\Delta\psi$, and roll accelerations are assumed negligible. These assumptions reduce the full lateral system (9.60) to the smaller 2×2 Dutch roll system:

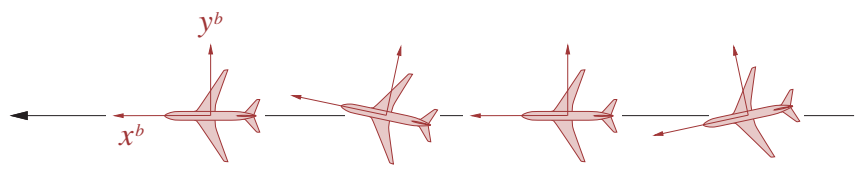

Figure 9.12: Assumed motion for the Dutch-roll approximation.

$$\begin{Bmatrix} \Delta\dot{v} \\ \Delta\dot{r} \end{Bmatrix} = \begin{bmatrix} Y_v & Y_r - u_0 \\ N_v & N_r \end{bmatrix} \begin{Bmatrix} \Delta v \\ \Delta r \end{Bmatrix} \tag{9.67}$$

This has the following two Dutch-roll eigenvalues.

$$\begin{aligned} \lambda_{1,2} &\equiv \sigma_{\mathrm{DR}} \pm \mathrm{i}\,\omega_{\mathrm{DR}} \\ \sigma_{\mathrm{DR}} &\simeq N_r + Y_v &&= \frac{QSb^2}{2I_z u_0}\left(C_{n_r} + \frac{I_z}{mb^2}C_{y_\beta}\right) \\ \omega_{\mathrm{DR}}^2 &\simeq u_0 N_v + Y_v N_r - N_v Y_r &&= \frac{QSb}{I_z}\left(C_{n_\beta} + \frac{\rho Sb}{4m}(C_{y_\beta}C_{n_r} - C_{n_\beta}C_{y_r})\right) \end{aligned} \tag{9.68}$$

It must be pointed out that the neglect of roll in the assumed Dutch-roll motion in Figure 9.12 makes the resulting eigenvalue expressions (9.68) somewhat questionable. In particular, a large wing dihedral is known to make the Dutch-roll mode of actual aircraft less damped or even unstable, but this is not represented in the approximate results above since the associated C_{ℓ_β} derivative does not appear. However, the eigenvalues of the full 4×4 lateral system (9.60) will capture this destabilizing effect of dihedral on the Dutch-roll mode.

9.13 Stability Derivative Estimation

In modern fight dynamics practice, the stability and control derivatives are most conveniently computed numerically. The simplest and most common such approach is to employ the Vortex Lattice method developed in Chapter 6, although more advanced panel or grid methods may also be used. However, the classical derivative estimation techniques, such as those described by Etkin [65] and Nelson [66], are still very useful to gain some insight into how the geometry of a conventional aircraft influences the derivatives and hence the aircraft's dynamical behavior. The derivatives are estimated in terms of the aircraft's geometric parameters listed in Table 9.1. The reference (wing) area, span, chord will be denoted by S, b, c as before.

Table 9.1: Parameters used to estimate stability derivatives of a conventional aircraft.

AR	aspect ratio ($= b^2/S$)
$S_{\rm h}$	horizontal tail area
$S_{\rm v}$	vertical tail area
$AR_{\rm h}$	horizontal tail's aspect ratio
$AR_{\rm v}$	vertical tail's aspect ratio
λ	wing taper ratio ($= c_{\rm tip}/c_{\rm root}$)
$\ell_{\rm h}$	distance from wing's aerodynamic center to horizontal tail
$\ell_{\rm v}$	distance from wing's aerodynamic center to vertical tail
$\ell_{\rm cg}$	distance from wing's aerodynamic center to mass centroid
Υ	effective wing dihedral angle (radians)
$\mathcal{V}_{\rm fuse}$	fuselage volume

9.13.1 Component derivatives

The estimation formulas are based on the lift-slope derivatives of the wing and tail surfaces, typically based on the lifting-line result (E.38) applied to each surface.

$$C_{L_\alpha} = \frac{c_{\ell_\alpha}}{1 + c_{\ell_\alpha}/(\pi AR)} \tag{9.69}$$

$$(C_{L_\alpha})_{\rm h} = \frac{(c_{\ell_\alpha})_{\rm h}}{1 + (c_{\ell_\alpha})_{\rm h}/(\pi AR_{\rm h})} \tag{9.70}$$

$$(C_{L_\alpha})_{\rm v} = \frac{(c_{\ell_\alpha})_{\rm v}}{1 + (c_{\ell_\alpha})_{\rm v}/(\pi AR_{\rm v})} \tag{9.71}$$

The lift-curve slopes c_{ℓ_α} of the wing and tail airfoils can be obtained from 2D airfoil experimental data or calculations. The thin-airfoil approximation $c_{\ell_\alpha} \simeq 2\pi$ is also commonly used. Compressibility effects can also be included here via the Prandtl-Glauert theory and sweep theory as in equations (8.87) and (8.103). It should also be noted that "endplate" effects on the vertical tail by the horizontal tail can significantly increase $(C_{L_\alpha})_{\rm v}$ from its estimate above.

The angle of attack seen by the horizontal tail is modified by its own induced downwash, and also by the downwash angle of the wing at the tail location. The latter is denoted by ε, and can be estimated as

$$\varepsilon \simeq k\,\frac{C_L}{\pi AR} \tag{9.72}$$

$$\frac{{\rm d}\varepsilon}{{\rm d}\alpha} \simeq k\,\frac{C_{L_\alpha}}{\pi AR} \tag{9.73}$$

$$k = 1 + \frac{1}{\sqrt{1+(\ell_{\rm h}/b)^2}}\left(\frac{1}{\pi\,\ell_{\rm h}/b} + 1\right) \tag{9.74}$$

where the factor k is obtained from a horseshoe-vortex model of the wing and its trailing vortex system. The effect of the downwash is to reduce the horizontal tail's effective lift-curve slope with respect to the aircraft angle of attack by the factor $1 - {\rm d}\varepsilon/{\rm d}\alpha$.

The drag coefficient's dependence on the angle of attack is commonly estimated assumed to be confined entirely to the induced drag component.

$$C_D = C_{D_p} + \frac{C_L^2}{\pi AR} \tag{9.75}$$

$$C_{D_\alpha} = \frac{2C_{L_0}\,C_{L_\alpha}}{\pi AR} \tag{9.76}$$

The fuselage's contribution to the overall lift is typically assumed to be captured by the center part of the wing "hidden" in the fuselage, which is already included in the C_{L_α} derivative. But a fuselage does typically have an additional significant pitching and yawing moment contributions. These can be estimated by the slender-body theory result (6.77), which has the following non-dimensional forms.

$$(C_{m_\alpha})_{\text{fuse}} = \frac{2\mathcal{V}_{\text{fuse}}}{Sc} \tag{9.77}$$

$$(C_{n_\beta})_{\text{fuse}} = \frac{-2\mathcal{V}_{\text{fuse}}}{Sb} \tag{9.78}$$

These contributions are destabilizing in pitch and yaw.

9.13.2 Longitudinal derivatives

The axial-force derivatives can now be estimated using the wing, horizontal-tail, and fuselage derivatives above. For the axial force we have

$$C_x = C_L \sin\alpha - C_D \cos\alpha \simeq C_L\,\alpha - C_D \tag{9.79}$$

$$C_{x_\alpha} = C_{L_0} - C_{D_\alpha} = C_{L_0}\left(1 - \frac{2C_{L_\alpha}}{\pi AR}\right) \tag{9.80}$$

$$C_{x_q} \simeq 0 \tag{9.81}$$

where small-angle approximations have been made. The normal force derivatives are estimated in the same manner,

$$C_z = -C_L \cos\alpha - C_D \sin\alpha \simeq -C_L - C_D\,\alpha \tag{9.82}$$

$$C_{z_\alpha} = -C_{L_\alpha} - C_{D_0} \tag{9.83}$$

$$C_{z_q} \simeq -2(C_{L_\alpha})_{\text{h}}\,V_{\text{h}} \tag{9.84}$$

$$V_{\text{h}} \equiv \frac{S_{\text{h}}\ell_{\text{h}}}{Sc} \tag{9.85}$$

with V_{h} denoting the horizontal tail volume coefficient.

The pitching moment is assumed to have wing, tail, and fuselage contributions.

$$C_{m_\alpha} = C_{L_\alpha}\frac{\ell_{\text{cg}}}{c} - (C_{L_\alpha})_{\text{h}}\,V_{\text{h}}\left(1 - \frac{\mathrm{d}\varepsilon}{\mathrm{d}\alpha}\right) + (C_{m_\alpha})_{\text{fuse}} \tag{9.86}$$

$$C_{m_q} = -2(C_{L_\alpha})_{\text{h}}\,V_{\text{h}}\,\frac{\ell_{\text{h}}}{c} + (C_{m_q})_{\text{fuse}} \tag{9.87}$$

The downwash factor $1-\mathrm{d}\varepsilon/\mathrm{d}\alpha$ is excluded from C_{m_q}. The reason is that this pitch-rate derivative dominates mainly in rapid pitching motions (e.g. in the short-period mode) where the wing's shed and trailing vorticity perturbations do not have sufficient time to convect to the tail to appreciably change the downwash there. However, the downwash factor can be added if the short-period mode is expected to be significantly slower than the wing–tail convection time. This will also give a smaller and hence more conservative estimate of pitch damping.

9.13.3 Lateral derivatives

The side force derivatives are estimated using the force derivative on the vertical tail.

$$C_{y_\beta} = -\frac{S_{\text{v}}}{S}\,(C_{L_\alpha})_{\text{v}} \tag{9.88}$$

$$C_{y_r} = -2\,V_{\text{v}}\,(C_{L_\alpha})_{\text{v}} \tag{9.89}$$

$$V_{\text{v}} \equiv \frac{S_{\text{v}}\ell_{\text{v}}}{Sb} \tag{9.90}$$

The vertical tail volume coefficient V_{v} naturally appears.

The rolling moment derivatives are obtained by using lifting-line theory applied to the wing sections, with the local velocities and resulting local angle of attack modified by the sideslip together with the wing dihedral, and by the roll and yaw rates. Integration of the local z-force perturbation across the span together with the local moment arm gives the rolling moment perturbation.

$$C_{\ell_\beta} = -C_{L_\alpha} \frac{\Upsilon}{6} \frac{1+2\lambda}{1+\lambda} \tag{9.91}$$

$$C_{\ell_p} = -C_{L_\alpha} \frac{1}{12} \frac{1+3\lambda}{1+\lambda} \tag{9.92}$$

$$C_{\ell_r} = \frac{C_{L_0}}{4} \tag{9.93}$$

The contributions of the tail surfaces are neglected here, but could be included with a more detailed analysis.

The yaw moment derivatives are obtained in a similar manner using the local z-force perturbations. The fuselage contribution is also included.

$$C_{n_\beta} = (C_{L_\alpha})_{\mathrm{v}}\, V_{\mathrm{v}} + (C_{n_\beta})_{\mathrm{fuse}} \tag{9.94}$$

$$C_{n_p} = -\frac{C_{L_0}}{8} \tag{9.95}$$

$$C_{n_r} = -2(C_{L_\alpha})_{\mathrm{v}}\, V_{\mathrm{v}}\, \frac{\ell_{\mathrm{v}}}{b} \tag{9.96}$$